

A Literature-Implied Negative Answer to the Higher Property (T) Converse

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2026-07-19

1 Status

This is a lightweight *literature-implied answer*, not an original proof. The source paper is

U. Bader and P. W. Nowak, *Cohomology of deformations*, arXiv:1401.5362.

In Section 5.1, “Higher property (T) and reduced cohomology”, Definition 5.1 defines property (T_n) for a group of type F_{n+1} by requiring that

$$H^{n+1}(\Gamma, \pi)$$

is reduced for every unitary representation π . Immediately after the definition, the paper asks whether the converse to the trivial implication is true: does this reducedness condition imply

$$H^{n+1}(\Gamma, \pi) = 0$$

for every unitary representation π ?

2 Supporting Literature

The supporting paper is U. Bader and R. Sauer, *Higher Kazhdan property and unitary cohomology of arithmetic groups*, arXiv:2308.06517v3. Its introduction explicitly says that the higher analogues of the Delorme–Guichardet equivalent criteria are not equivalent. Its Definition 1.1 uses different notation: $[T_n]$ is vanishing of H^j for every unitary representation and $1 \leq j \leq n$, while (T_n) is the corresponding vanishing condition for unitary representations without invariant vectors.

The decisive passage is Remark 3.10. It states that, for $n > 1$, having

$$H^k(\Gamma, U) \text{ reduced for every } k \leq n \text{ and every unitary } U$$

does not guarantee Bader–Sauer’s vanishing property (T_n) , and gives $\mathrm{SL}_{n+1}(\mathbb{Q}_p)$ and its lattices as examples with reduced cohomology groups in all degrees.

3 Identification

Take $m \geq 1$ in the Bader–Nowak question and set $k = m + 1 \geq 2$. Let Γ be a suitable lattice in $\mathrm{SL}_{k+1}(\mathbb{Q}_p)$, for instance a torsion-free cocompact lattice when one wants a discrete group with a finite $K(\Gamma, 1)$. Bader–Sauer’s Remark 3.10 gives that

$$H^j(\Gamma, U)$$

is reduced for every degree j and every unitary U . In particular, Γ satisfies the Bader–Nowak property (T_m) , since the relevant degree is $m + 1 = k$.

On the other hand, the same Bader–Sauer discussion says these examples do not satisfy the corresponding degree- k vanishing property. Garland’s theorem, as stated in the supporting paper for $\mathrm{SL}_r(\mathbb{Q}_p)$, gives vanishing through degree $r - 2$. For $r = k + 1$, this is vanishing through degree $k - 1$, so the first possible failure of vanishing is exactly degree k . Thus there is a unitary representation U with

$$H^k(\Gamma, U) \neq 0,$$

while $H^k(\Gamma, U)$ is reduced.

This is precisely a counterexample to the source converse for $m = k - 1$: reducedness of H^{m+1} for every unitary representation does not force vanishing of H^{m+1} for every unitary representation.

4 Scope

The implication is negative in every genuinely higher degree $m \geq 1$. The case $m = 0$ is different: it is the ordinary Delorme–Guichardet theorem for Kazhdan’s property (T), and remains positive.

The supporting authors do not present the passage as a direct answer to the question in arXiv:1401.5362. This packet records the consequence after matching definitions and the degree shift.

5 References

- U. Bader and P. W. Nowak, *Cohomology of deformations*, arXiv:1401.5362.
- U. Bader and R. Sauer, *Higher Kazhdan property and unitary cohomology of arithmetic groups*, arXiv:2308.06517v3.
- H. Garland, *p-adic curvature and the cohomology of discrete subgroups of p-adic groups*, *Annals of Mathematics* 97 (1973), as quoted in the supporting paper.